

# Factory Energy Analyser: Energy Performance, Waste Screening and Savings Plan

T15 – Factory Energy Analyser | No-code apps & dashboards

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## 1. Executive Summary

This report analyses 365 days of plant energy data (01-Apr-2025 to 31-Mar-2026) from the Factory Energy Analyser workbook. It measures energy performance, screens non-production days for unusual consumption, and sets out a 30/60/90-day energy plan. The plant used **4,489,639 kWh** (4,014,871 kWh grid and 474,768 kWh solar), a **solar share of 10.57%**. On the 313 production days, energy intensity was **7.55 kWh per unit** (4,271,630 kWh over 566,026 units). The 52 non-production days still used **218,009 kWh (4.86% of total energy)**, and three of them (11-May-25, 18-May-25, 22-Jun-25) exceeded the 95th-percentile screening threshold of 4,687.7 kWh. On production days, and holding units produced constant, each additional °C was **associated** with about 39.8 kWh more daily energy (multiple-regression  $R^2 \approx 0.765$ ).

**Main recommendation:** investigate non-production energy use first, then track production-day intensity, evaluate solar expansion and temperature-related loads, and use breakdown data only as an investigation signal. Five savings scenarios (₹8/kWh, illustrative) are presented as hypotheses that need validation; they are not guaranteed and must not be added together. All data are synthetic.

|                                      |                                                  |                              |                                                        |                                                      |                                                                  |
|--------------------------------------|--------------------------------------------------|------------------------------|--------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------|
| <b>4,489,639 kWh</b><br>Total energy | <b>7.55 kWh/unit</b><br>Production-day intensity | <b>10.57%</b><br>Solar share | <b>218,009 kWh</b><br>Non-production energy<br>(4.86%) | <b>3 days</b><br>Statistically flagged<br>(95th pct) | <b>+39.8 kWh/°C</b><br>Temp. association, units<br>held constant |
|--------------------------------------|--------------------------------------------------|------------------------------|--------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------|

## 2. Project Objective and Data

The Factory Energy Analyser turns daily plant records into a decision-support dashboard that shows energy use per unit produced, solar share and temperature effect, finds high-consumption non-production days, and links the findings to an energy plan. The dataset (sheet “Daily Energy”) holds **365 daily records from 01-Apr-2025 to 31-Mar-2026**. The variables used are production\_day, units\_produced, avg\_temp\_c, grid\_kwh, solar\_kwh, peak\_demand\_kw and machine\_breakdowns. The dataset is stated in the Task Brief to be synthetic and created for teaching.

**Data-quality validation (Data Validation sheet, all checks pass):** 365 records; 0 missing values across all 10 columns; 0 duplicate dates; continuous date range with no gaps; only valid Yes/No production\_day values (313 Yes, 52 No); no negative operational values; zero units on all non-production days and non-zero units on all production days. Validating first matters because every later KPI depends on a clean production-day flag.

## 3. Dashboard and KPI Findings

The Dashboard sheet presents six KPI cards, monthly grid and solar energy, monthly intensity, monthly solar share, a temperature-versus-energy chart, the flagged-day table, business insights, the savings scenarios and the 30/60/90-day plan. Its purpose is to monitor energy performance, show where investigation is needed, track solar contribution, reveal monthly patterns and support management decisions. Build file: FINAL\_AI\_Done.xlsx (Dashboard sheet); screen recording/link, if required: [TO BE ADDED].

| KPI                                    | Value                | Basis                                                          |
|----------------------------------------|----------------------|----------------------------------------------------------------|
| Total energy                           | 4,489,639 kWh        | Grid 4,014,871 + solar 474,768 kWh, all 365 days               |
| Solar share                            | 10.57%               | Annual solar ÷ annual total energy (not an average of daily %) |
| Total units produced                   | 566,026 units        | All output occurs on production days                           |
| Production / non-production days       | 313 / 52             | production_day = Yes / No                                      |
| Production-day energy                  | 4,271,630 kWh        | Total energy on production days only                           |
| <b>Production-day energy intensity</b> | <b>7.55 kWh/unit</b> | Production-day energy ÷ production-day units                   |
| Non-production energy                  | 218,009 kWh (4.86%)  | Energy on the 52 non-production days ÷ total energy            |
| Maximum peak demand                    | 520 kW               | Maximum of daily peaks (never summed)                          |
| Total machine breakdowns               | 364                  | Sum of machine_breakdowns                                      |

**Why intensity uses production days only.** Intensity is production-day energy divided by production-day units. Dividing total annual energy by total units would load the 218,009 kWh used on days with zero output onto the production figure (arithmetically about 7.93 kWh/unit), mixing two different questions. Non-production energy is therefore tracked as its

own KPI. The 7.55 kWh/unit figure is a plant-level benchmark for producing days. It includes fixed loads and any temperature effect, and the dataset has no product-mix or process-level detail, so it should be read as a signal for where to look, not as a verdict on efficiency.

#### 4. Monthly Analysis

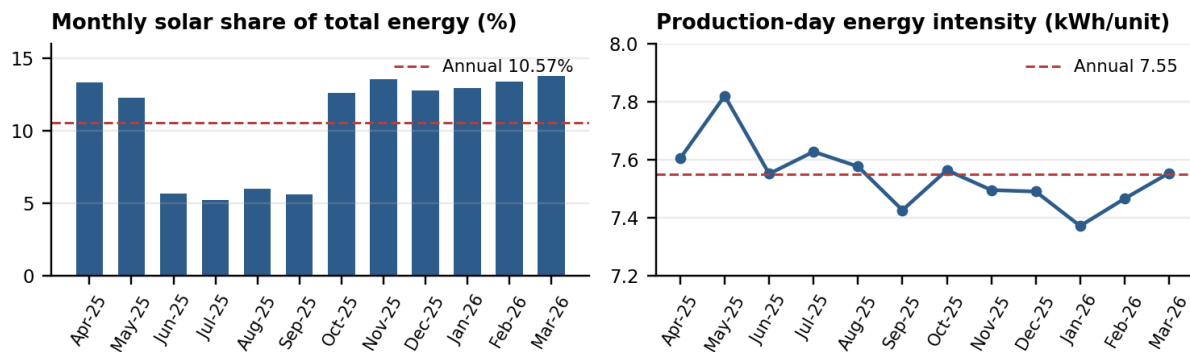


Figure 1. Values from the Monthly Analysis sheet. Monthly intensity = aggregated production-day energy ÷ aggregated production-day units (ratio of sums), not an average of daily ratios.

- **Total energy:** from 339,075 kWh (Feb-26, a 28-day month) to 403,243 kWh (May-25), so monthly totals partly reflect month length.
- **Intensity:** a narrow band of 7.37 kWh/unit (Jan-26) to 7.82 kWh/unit (May-25) around the 7.55 annual figure; May-25 is also the month with the highest total energy.
- **Solar share:** two distinct levels. Jun–Sep 2025 ran at 5.20%–5.99% (Jul-25 lowest, with 20,161 kWh solar), against 12.29%–13.76% in the other eight months (Mar-26 highest, with 52,322 kWh solar). The dataset does not explain the drop.
- **Non-production energy:** 15,656 kWh (Sep-25) to 22,519 kWh (Jun-25, five non-production days and the month of the highest flagged day).
- **Temperature and peak demand:** average production-day temperature ranged from 21.0°C (Jan-26) to 32.9°C (Jul-25). Monthly peak demand stayed between 493 kW (Dec-25) and 520 kW (Feb-26). Monthly breakdowns ranged from 24 (Aug-25) to 37 (Nov-25).

#### 5. Wasteful-Day Screening

**Method.** Only the 52 non-production days were screened. The threshold is the 95th percentile of their total daily kWh (inclusive method, PERCENTILE.INC), which is **4,687.7 kWh**; days strictly above it are flagged. Non-production days averaged 4,192.5 kWh (median 4,230.5; maximum 5,179), against a production-day average of 13,647.4 kWh. Three days are flagged as **potentially wasteful**:

| Date (Sunday) | Total kWh | Grid kWh | Solar kWh | Peak kW | Avg °C | kWh above threshold | % above non-prod. avg |
|---------------|-----------|----------|-----------|---------|--------|---------------------|-----------------------|
| 11-May-2025   | 4,917     | 3,054    | 1,863     | 448     | 30.3   | 229.3               | 17.3%                 |
| 18-May-2025   | 4,790     | 3,251    | 1,539     | 381     | 31.1   | 102.3               | 14.3%                 |
| 22-Jun-2025   | 5,179     | 4,036    | 1,143     | 432     | 34.3   | 491.3               | 23.5%                 |

These are **statistically flagged non-production days, not confirmed waste**. A percentile screen flags roughly the top 5% of days by construction, and the fourth-highest day (25-May-25, 4,604 kWh) sits just below the line. The data records two machine breakdowns on each flagged day, but the dataset has no cause field, so any link is unestablished. The next step is to investigate possible standby loads, shutdown procedures, equipment left running, abnormal operating conditions, and maintenance or other operational activity. These are areas to check, not identified causes.

#### 6. Temperature–Energy Analysis

The primary model uses production days only (313 observations): **total daily kWh = intercept +  $b_1 \times$  units produced +  $b_2 \times$  average temperature**. Units produced carries 5.57 kWh per unit and the intercept is 2,497 kWh. Holding units constant, the temperature coefficient is **+39.8 kWh per °C (95% CI 28.6 to 50.9)**, with  $R^2 \approx 0.765$  for the model as a whole.

The simple temperature-only regression is a different model: slope 55.0 kWh/°C,  $R^2 \approx 0.072$ . It ignores production volume, which drives most day-to-day variation in energy, so temperature alone explains little. The 0.765 belongs to the units-plus-temperature model and should not be read as temperature's own explanatory power; the two slopes are not directly comparable because only one controls for units. **After controlling for production volume, higher temperature was associated with higher daily energy consumption.** This does not show that temperature causes the extra energy. It

suggests that cooling/HVAC or other temperature-related loads may be worth investigating, although the data has no dedicated HVAC field and omits other operating or environmental variables.

## 7. Five Savings Scenarios (Illustrative)

Savings are valued at an **illustrative tariff of ₹8/kWh** (₹ impact = kWh × tariff). They are scenario estimates, **not guaranteed savings and not additive**, because initiatives overlap; the workbook deliberately shows no total.

| # | Initiative                                  | kWh opportunity | Illustrative ₹ impact | % of annual energy | Baseline, assumption and validation needed                                                                                                                          |
|---|---------------------------------------------|-----------------|-----------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Non-production shutdown / standby reduction | 130,805         | ₹1,046,443            | 2.91%              | Equivalent to 60.0% of non-production energy (218,009 kWh)†. Needs a shutdown/standby audit.                                                                        |
| 2 | Production efficiency improvement           | 128,149         | ₹1,025,191            | 2.85%              | Equivalent to 3.0% of production-day energy (4,271,630 kWh)†. Needs process-level validation.                                                                       |
| 3 | Solar generation expansion                  | 94,954          | ₹759,629              | 2.11%              | Equivalent to 20.0% of current solar output (474,768 kWh)†. Additional generation / grid displacement, not lower underlying demand; capex and payback not included. |
| 4 | Breakdown-linked controls                   | 42,716          | ₹341,730              | 0.95%              | Equivalent to 1.0% of production-day energy†. Hypothetical: energy caused by breakdowns is not measured.                                                            |
| 5 | Cooling / HVAC optimisation                 | 7,789           | ₹62,312               | 0.17%              | Baseline [TO BE ADDED]. Scenario-based: no dedicated HVAC energy field exists.                                                                                      |

† The workbook records the kWh estimates but not the percentage assumptions behind them; the percentages shown are the ratios of each kWh figure to the stated baseline, and the original assumptions should be confirmed [TO BE ADDED]. kWh and ₹ rounded to whole numbers. The five figures are not ranked by certainty and should be validated technically, financially and for overlap before any commitment.

## 8. Energy Plan (One-Page Summary)

**Situation.** The plant used 4,489,639 kWh at a 10.57% solar share and 7.55 kWh/unit production-day intensity. The 52 non-production days used 218,009 kWh, and 3 were statistically flagged. Temperature is associated with energy use after controlling for units, but cause is not established.

| Horizon | Focus               | Action                                                                                                 | Measure of success                                      |
|---------|---------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| 30 days | Non-production days | Investigate the three statistically flagged non-production days and establish shutdown/standby checks. | Cause identified for each flagged day; checklist in use |
| 60 days | Energy intensity    | Track production-day energy intensity and investigate high-intensity periods.                          | Monthly kWh/unit against the 7.55 annual baseline       |
| 90 days | Solar and cooling   | Evaluate solar expansion and cooling/HVAC optimisation opportunities.                                  | Feasibility and payback review completed                |

**KPIs to monitor:** kWh/unit; non-production kWh/day; solar share; peak demand; machine breakdowns; validated rupee savings. **Financial and validation considerations:** all ₹ values use the illustrative ₹8/kWh tariff; solar needs capex and payback analysis; savings should be counted only after validation and overlap checks. Action owners and numeric targets: [TO BE ADDED] (to be agreed by management).

## 9. Recommendation

Based only on the evidence in the workbook, management should act in this order:

1. **Investigate non-production energy first.** 218,009 kWh was used on days with no output, and three days were statistically flagged. Begin with the 11-May-25, 18-May-25 and 22-Jun-25 records and introduce shutdown/standby checks.
2. **Improve production energy intensity.** Track monthly kWh/unit against the 7.55 baseline and review high-intensity periods such as May-25 (7.82 kWh/unit).
3. **Evaluate solar expansion.** Solar supplied 10.57% overall but only 5.2%–6.0% in Jun–Sep 2025; a feasibility and payback review is needed before any commitment.
4. **Investigate temperature-related energy use.** Check whether cooling/HVAC or similar loads explain the +39.8 kWh/°C association, treating it as an association.
5. **Use breakdown data as an investigation signal.** Breakdown counts (364 in the year) can prompt checks, but the data does not quantify energy caused by breakdowns.

## 10. Verification and Quality Assurance

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The QA Checks sheet reports “**ALL 50 CHECKS PASS**” (50 of 50). Numerical checks reconcile total, grid and solar energy; monthly totals to annual totals; solar share; units; production-day energy; production-day intensity (including a direct recomputation of  $4,271,630 \div 566,026$ ); non-production energy; peak demand; the waste-screening threshold, count, dates and kWh values; the temperature coefficient, confidence interval and both  $R^2$  values; the dashboard KPI cards; and the five ₹ savings figures. The Data Validation sheet adds its own integrity checks. Calculations were independently checked against the supplied workbook; no external validation was carried out.

## 11. AI Use and Responsible AI

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The AI Use Log contains **15 ChatGPT interactions** (the sheet's first row is a template example and is not counted), covering dataset inspection, KPI definition, waste screening, temperature analysis, monthly analysis, savings scenarios and dashboard design. Three improvement rounds (entries 9 to 11) reviewed and revised the dashboard: a critique against the brief and rubric, a numerical QA review that separated the two  $R^2$  values and fixed the KPI definitions, and a business-usability review. Further entries reviewed the Energy Plan and Savings Table, the final KPIs, the wasteful-day method and the end-to-end workbook. AI supported analytical structuring, interpretation, dashboard design, statistical reasoning, scenario development and QA. Several outputs were only partly kept or changed, for example the waste screen was moved to the 95th percentile.

AI numbers were independently checked against the workbook, and AI-generated assumptions and recommendations were treated as hypotheses or scenarios, not measured facts. The completed log carries an evidence link for each entry. The data is synthetic teaching data with no confidential business or personal information.

## 12. Limitations

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- The dataset is synthetic teaching data and does not establish causality.
- Wasteful-day flags are statistical screens, not confirmed operational waste; the data has no cause field.
- Savings figures are scenario estimates; ₹8/kWh is an illustrative tariff and initiatives may overlap, so they should not be added.
- Breakdown data does not directly quantify energy caused by breakdowns, and there is no dedicated HVAC energy field.
- Solar expansion estimates exclude capex and payback.
- The temperature analysis omits other possible operational and environmental variables.

## 13. Reflection

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This project showed me that a dashboard is only as reliable as the checks behind it. Validating the data first (record counts, duplicates, date continuity and the production-day flag) mattered because every KPI depends on that flag. Defining intensity as production-day energy over production-day units changed the business reading, since the non-production load is a separate issue, not a distortion hidden in one average. I also learned the difference between screening and proof. The 95th-percentile screen narrows attention to three days, but it cannot say why they were high, and the same discipline applies to temperature: the regression shows an association after controlling for units, not a cause.

Iterative prompting improved the work. Reviewing my own dashboard and plan in later rounds exposed weak wording, the confusion between the two  $R^2$  values and over-dense layouts. It also showed that AI outputs need verification: I treated suggestions as hypotheses and checked figures against the workbook. With more operational data, such as sub-metered HVAC and standby loads, shutdown logs, breakdown downtime, product mix and tariff and capex details, I would test the causes behind the flagged days and the temperature association and replace the illustrative scenarios with validated estimates.